

MEDFORD, OR, USA

WMO: 725970

Lat: 42.3769N

Lon: 122.8822W

Elev: 1309

StdP: 14.01

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 24225

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.9	26.7	16.6	13.3	37.3	20.6	16.2	37.9	25.3	52.9	21.1	54.5	1.1	0	0.481

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	31.3	98.9	66.8	95.5	65.6	92.3	64.4	68.7	93.9	67.2	91.4	65.8	88.8	8.8	290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.1	81.4	73.9	58.1	75.8	73.8	56.5	71.5	72.9	33.5	93.8	32.2	91.4	31.1	88.5	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.3	15.4	12.4	DB	18.5	105.1	5.3	3.2	14.7	107.5	11.6	109.4	8.6	111.2	4.8	113.6	(4)
(5)			WB	17.5	71.7	5.1	1.6	13.8	72.9	10.8	73.8	8.0	74.7	4.3	75.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	56.0	40.5	44.1	48.2	52.7	60.4	67.0	75.7	74.5	67.6	55.8	45.4	39.4		
	DBStd	13.85	5.47	5.34	5.67	6.58	7.08	6.76	5.63	5.35	6.47	6.15	5.87	6.37		
	HDD50	1133	297	176	104	43	4	0	0	0	0	17	159	333		
	HDD65	4195	759	586	520	371	180	54	2	3	48	292	587	793		
	CDD50	3333	3	10	48	126	326	511	797	760	529	197	22	4		
	CDD65	922	0	0	0	4	37	115	334	298	127	7	0	0		
	CDH74	12575	0	0	8	149	719	1694	4275	3709	1815	206	0	0		
	CDH80	6349	0	0	0	37	275	761	2364	1989	873	50	0	0		
	WSAvg	3.9	2.9	3.6	4.2	4.5	4.9	5.2	5.0	4.4	3.5	2.9	2.8	3.0		
	PrecAvg	18.7	2.7	2.0	1.9	1.3	1.1	0.6	0.2	0.4	0.7	1.4	3.0	3.4		
(15) Precipitation	PrecMax	31.4	6.2	5.7	4.4	3.6	4.3	2.6	1.6	2.8	4.2	6.3	7.7	12.7		
	PrecMin	9.0	0.2	0.2	0.3	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.4		
	PrecStd	4.6	1.6	1.3	1.0	0.8	0.8	0.6	0.4	0.6	0.8	1.3	1.9	2.5		
	PrecStd	4.6	1.6	1.3	1.0	0.8	0.8	0.6	0.4	0.6	0.8	1.3	1.9	2.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	66.8	74.7	85.1	93.4	98.9	104.1	103.5	98.3	86.0	67.9	60.3		
		MCWB	49.5	52.5	55.4	59.6	63.7	67.4	68.5	67.4	63.9	60.1	53.6	50.3		
	2%	DB	56.3	61.8	69.5	78.3	87.4	93.0	99.7	98.0	93.9	79.7	62.7	55.7		
		MCWB	47.6	49.7	52.8	57.7	62.2	65.1	67.8	65.9	63.1	57.7	52.0	48.1		
	5%	DB	52.4	57.9	64.7	72.4	82.1	88.4	96.1	94.4	89.7	74.7	58.6	52.0		
		MCWB	46.3	47.6	51.3	54.9	59.8	63.0	66.6	64.9	62.2	56.2	50.4	47.3		
	10%	DB	49.6	54.4	60.3	66.7	76.8	83.9	92.3	91.1	85.1	69.8	55.0	48.8		
		MCWB	45.0	46.3	49.4	52.6	57.8	61.3	65.4	63.8	60.6	54.5	49.2	45.2		
	0.4%	WB	52.2	54.6	57.2	60.9	66.1	69.3	71.6	69.4	67.3	61.8	57.6	53.0		
		MCDB	55.7	64.1	71.2	80.7	88.6	93.9	97.6	96.4	90.4	80.3	64.3	56.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	49.6	51.6	54.5	58.4	63.6	66.7	69.4	67.7	65.0	59.6	54.3	50.0		
		MCDB	54.5	58.2	65.4	75.8	83.5	89.4	95.0	93.6	87.5	75.5	59.3	54.0		
	5%	WB	47.5	49.4	52.5	55.9	61.3	64.3	67.9	66.3	63.4	57.6	52.1	47.7		
		MCDB	51.2	56.0	62.3	70.0	78.7	85.4	92.3	90.6	85.4	70.6	56.6	51.0		
	10%	WB	45.6	47.2	50.4	53.4	59.0	62.3	66.3	64.9	61.8	55.5	50.0	45.7		
		MCDB	49.2	53.1	58.8	64.9	74.1	81.2	89.5	87.6	82.6	66.8	54.5	48.7		
	5% DB	MDBR	14.6	18.7	21.0	22.6	25.4	28.0	31.3	31.7	31.3	26.0	16.3	12.7		
		MCDBR	19.9	25.6	31.0	34.7	35.6	35.6	36.0	36.2	39.1	35.5	23.3	16.9		
		MCWBR	12.5	14.9	16.3	16.2	14.2	13.0	12.2	11.4	13.9	16.1	13.5	10.5		
		MCWBR	16.0	21.7	26.8	31.4	32.4	33.7	33.5	33.7	35.5	30.2	18.3	13.7		
(35) Mean Daily Temperature Range	5% WB	MCWBR	10.8	13.2	14.3	15.2	13.3	12.6	12.0	11.1	13.3	14.3	11.3	9.0		
		MCWBR	10.8	13.2	14.3	15.2	13.3	12.6	12.0	11.1	13.3	14.3	11.3	9.0		
	Clear-Sky Solar Irradiance	taub	0.305	0.307	0.310	0.332	0.340	0.337	0.343	0.340	0.320	0.317	0.302	0.311		
		taud	2.551	2.559	2.571	2.486	2.498	2.530	2.527	2.524	2.562	2.578	2.587	2.527		
(42) All-Sky Solar Radiation	RadAvg	Ebn at Noon	269	287	299	297	296	296	293	291	291	279	269	255		
		Edh at Noon	22	25	28	33	33	32	32	31	28	24	21	21		
(44) All-Sky Solar Radiation	RadStd	RadAvg	547	817	1173	1617	2067	2392	2521	2200	1706	1092	611	438		
		RadStd	82	114	123	155	170	143	90	103	98	108	76	82		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+189	+126
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+7

Nomenclature: See separate page